

Principal Component Analysis (PCA)

Lecture Notes on Dimensionality Reduction in Machine Learning

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Abstract

Principal Component Analysis (PCA) is a fundamental dimensionality reduction technique in machine learning and data science. These comprehensive lecture notes provide a detailed mathematical foundation, step-by-step algorithmic implementation, worked examples, and practical applications of PCA. The material is designed to facilitate understanding of complex concepts through clear explanations, visual interpretations, and practice problems.

1 Introduction to PCA

1.1 What is Principal Component Analysis?

Key Concept

Principal Component Analysis (PCA) is a **linear dimensionality reduction technique** that transforms high-dimensional data into a lower-dimensional space while preserving the maximum variance in the data. It identifies orthogonal directions (principal components) along which the data varies the most.

1.2 Why Use PCA?

PCA serves multiple crucial purposes in data analysis and machine learning:

- **Dimensionality Reduction:** Reduces the number of features while retaining essential information
- **Visualization:** Projects high-dimensional data (e.g., 10+ features) to 2D or 3D for visualization
- **Noise Reduction:** Eliminates components with low variance (often representing noise)
- **Computational Efficiency:** Reduces training time and memory requirements
- **Multicollinearity Removal:** Creates uncorrelated predictors for regression models
- **Feature Extraction:** Identifies underlying patterns in the data

1.3 Core Mathematical Intuition

The geometric interpretation of PCA involves:

1. **Centering:** Shifting the origin to the mean of the data
2. **Rotation:** Finding new orthogonal axes that maximize variance
3. **Projection:** Mapping data points onto the new axes

Mathematically, PCA finds eigenvectors of the covariance matrix that represent directions of maximum variance, with eigenvalues representing the magnitude of variance along each direction.

2 Mathematical Foundation

2.1 Key Mathematical Concepts

2.1.1 Covariance Matrix

The covariance matrix S measures how features vary together:

$$\text{Cov}(X_i, X_j) = \frac{1}{N-1} \sum_{k=1}^N (x_{ki} - \bar{X}_i)(x_{kj} - \bar{X}_j)$$

For an n -dimensional dataset:

$$\mathbf{S} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n) \end{bmatrix}$$

2.1.2 Eigenvalues and Eigenvectors

Eigenvalues (λ) and eigenvectors ($\mathbf{v}$) satisfy:

$$\mathbf{S}\mathbf{v} = \lambda\mathbf{v}$$

- **Eigenvalue** λ : Amount of variance captured
- **Eigenvector** $\mathbf{v}$: Direction of principal component

2.1.3 Variance Explained

The proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_{j=1}^n \lambda_j}$$

3 The PCA Algorithm: Step-by-Step

3.1 Formal Algorithm Description

Given a dataset with N observations and n features:

Step 1: Standardize the Data (if necessary):

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}$$

Step 2: Compute Mean Vector:

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$$

Step 3: Center the Data:

$$\mathbf{X}_{\text{centered}} = \mathbf{X} - \bar{\mathbf{x}}$$

Step 4: Compute Covariance Matrix:

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}_{\text{centered}}^T \mathbf{X}_{\text{centered}}$$

Step 5: Compute Eigenvalues and Eigenvectors:

$$\det(\mathbf{S} - \lambda\mathbf{I}) = 0$$

Solve for eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Step 6: Normalize Eigenvectors:

$$\mathbf{e}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$$

Step 7: Select Principal Components: Choose k components based on variance explained threshold

Step 8: Project Data:

$$\mathbf{Z} = \mathbf{X}_{\text{centered}} \mathbf{E}_k$$

where $\mathbf{E}_k = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k]$

3.2 Choosing the Number of Components

- **Scree Plot Method:** Plot eigenvalues vs. component number, look for "elbow"
- **Cumulative Variance Threshold:** Choose k such that:

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^n \lambda_i} \geq \tau$$

where τ is typically 0.95 or 0.90

- **Kaiser Criterion:** Keep components with eigenvalues > 1 (for standardized data)

4 Detailed Worked Examples

4.1 Example 1: 2D to 1D Reduction (4 Points)

Example

Dataset: $(4, 11), (8, 4), (13, 5), (7, 14)$

Goal: Reduce from 2D to 1D

4.1.1 Step 1: Compute Means

$$\bar{X}_1 = \frac{4 + 8 + 13 + 7}{4} = 8, \quad \bar{X}_2 = \frac{11 + 4 + 5 + 14}{4} = 8.5$$

4.1.2 Step 2: Center the Data

$$\mathbf{X}_{\text{centered}} = \begin{bmatrix} -4 & 2.5 \\ 0 & -4.5 \\ 5 & -3.5 \\ -1 & 5.5 \end{bmatrix}$$

4.1.3 Step 3: Compute Covariance Matrix

$$\mathbf{S} = \frac{1}{3} \mathbf{X}_{\text{centered}}^T \mathbf{X}_{\text{centered}} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

4.1.4 Step 4: Compute Eigenvalues

$$\det \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} = 0 \Rightarrow \lambda^2 - 37\lambda + 201 = 0$$

$$\lambda_1 \approx 30.3849, \quad \lambda_2 \approx 6.6151$$

4.1.5 Step 5: Compute Eigenvectors

For $\lambda_1 = 30.3849$:

$$\begin{bmatrix} -16.3849 & -11 \\ -11 & -7.3849 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \mathbf{0}$$
$$\mathbf{v}_1 \approx \begin{bmatrix} 1 \\ -1.4895 \end{bmatrix}, \quad \mathbf{e}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \approx \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

4.1.6 Step 6: Project Data

$$\mathbf{Z} = \mathbf{X}_{\text{centered}} \mathbf{e}_1 \approx \begin{bmatrix} -4.3054 \\ 3.7365 \\ 6.2192 \\ -5.6503 \end{bmatrix}$$

4.2 Example 2: 2D to 1D Reduction (10 Points)

Example

Dataset: (2.5, 2.4), (0.5, 0.7), (2.2, 2.9), (1.9, 2.2), (3.1, 3.0),
(2.3, 2.7), (2.0, 1.6), (1.0, 1.1), (1.5, 1.6), (1.1, 0.9)

4.2.1 Step 1: Compute Means

$$\bar{X} = 1.81, \quad \bar{Y} = 1.91$$

4.2.2 Step 2: Compute Covariance Matrix

$$\mathbf{S} = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}$$

4.2.3 Step 3: Compute Eigenvalues

$$\lambda^2 - 1.3332\lambda + 0.0629 = 0$$
$$\lambda_1 = 1.28425, \quad \lambda_2 = 0.04895$$

4.2.4 Step 4: Compute Eigenvectors

For $\lambda_1 = 1.28425$:

$$\mathbf{e}_1 \approx \begin{bmatrix} 0.677 \\ 0.735 \end{bmatrix}$$

4.2.5 Step 5: Project Data

For point (2.5, 2.4):

$$(2.5 - 1.81, 2.4 - 1.91) \cdot \begin{bmatrix} 0.677 \\ 0.735 \end{bmatrix} \approx 0.8273$$

4.3 Example 3: Simple Linear Relationship

Example

Dataset: (1, 1), (3, 3), (5, 5)

4.3.1 Step 1: Compute Means

$$\bar{X} = 3, \quad \bar{Y} = 3$$

4.3.2 Step 2: Center the Data

$$\mathbf{X}_{\text{centered}} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

4.3.3 Step 3: Compute Covariance Matrix

$$\mathbf{S} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

4.3.4 Step 4: Compute Eigenvalues

$$\lambda^2 - 8\lambda = 0 \Rightarrow \lambda_1 = 8, \quad \lambda_2 = 0$$

4.3.5 Step 5: Compute Eigenvectors

For $\lambda_1 = 8$:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

4.3.6 Interpretation

The second eigenvalue is 0, indicating all variance is captured by PC1. The data lies perfectly on the line $y = x$.

5 Applications and Interpretation

5.1 Practical Applications

- **Image Processing:** Face recognition (Eigenfaces)
- **Finance:** Portfolio optimization, risk management
- **Genetics:** Gene expression analysis
- **Signal Processing:** Noise reduction
- **Natural Language Processing:** Document clustering
- **Computer Vision:** Feature extraction
- **Neuroscience:** fMRI data analysis
- **Chemometrics:** Spectral data analysis

5.2 Geometric Interpretation

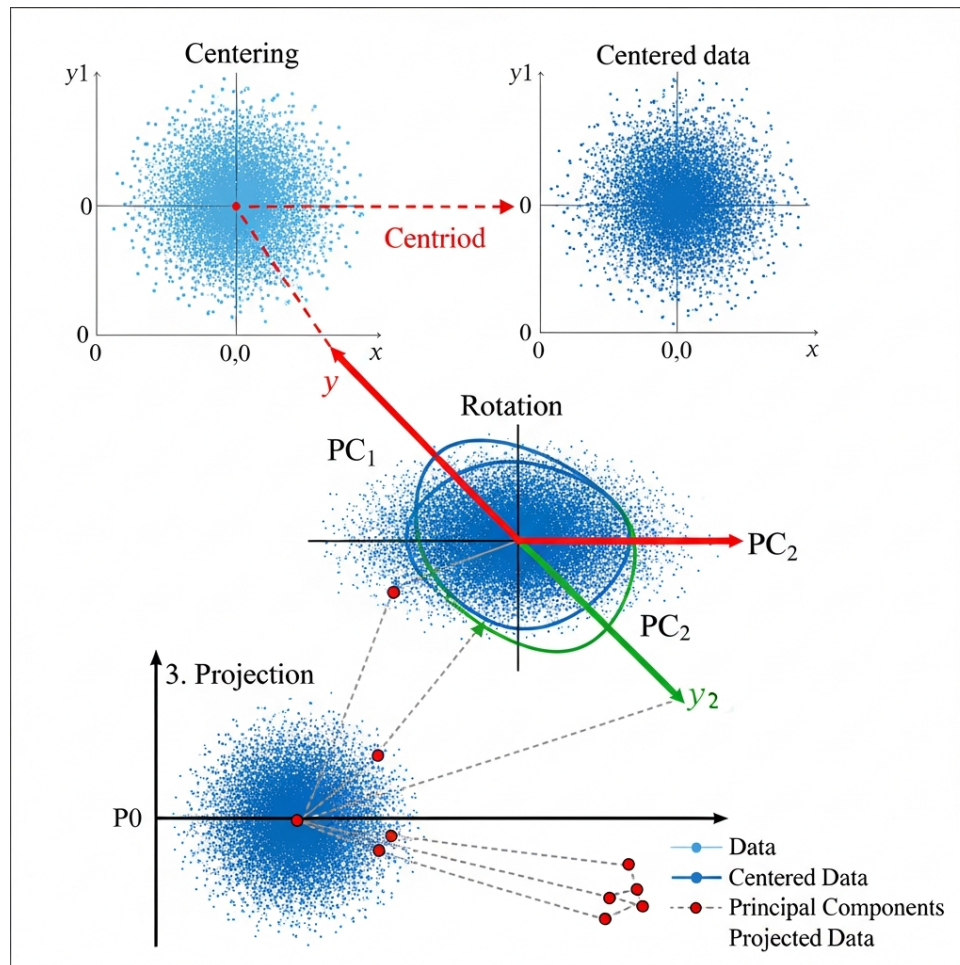


Figure 1: Geometric interpretation of PCA: centering, rotation, and projection

The PCA transformation can be visualized as:

1. **Translation:** Center data at origin
2. **Rotation:** Align axes with directions of maximum variance
3. **Scaling:** Scale axes by standard deviation (eigenvalues)
4. **Projection:** Drop dimensions with low variance

5.3 Variance Explained Analysis

Table 1: Variance explained by principal components

Component	Eigenvalue	Variance Explained
PC1	30.3849	82.1%
PC2	6.6151	17.9%

5.4 Limitations of PCA

- **Linearity Assumption:** PCA captures only linear relationships

- **Scale Sensitivity:** Results depend on feature scaling
- **Outlier Sensitivity:** Outliers can distort principal components
- **Interpretability:** Principal components are linear combinations, not original features
- **Information Loss:** Variance explained information content

6 Exercise Problems

6.1 Problem 1: Covariance Calculation

Problem

Given two features:

$$X : [1, 2, 3, 4, 5], \quad Y : [2, 4, 6, 8, 10]$$

Compute the covariance matrix.

6.2 Problem 2: Eigenvalue Computation

Problem

For the covariance matrix:

$$S = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

6.3 Problem 3: Dimensionality Decision

Problem

You performed PCA on a 10-feature dataset. Eigenvalues:

$$[4.5, 2.3, 1.8, 0.9, 0.5, 0.3, 0.2, 0.1, 0.05, 0.05]$$

How many components would you retain to explain at least 90% of the variance?

6.4 Problem 4: PCA Projection

Problem

Given centered data point $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and principal component $e = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, compute the projection.

6.5 Problem 5: Real-World Application

Problem

You have a dataset of house prices with features: size (sq ft), number of bedrooms, age (years), distance to city center (miles), and school rating (1-10). Describe how you would apply PCA and what insights you might gain.

7 Solutions to Exercise Problems

7.1 Solution to Problem 1

Means: $\bar{X} = 3, \bar{Y} = 6$

Centered data: $X' : [-2, -1, 0, 1, 2], Y' : [-4, -2, 0, 2, 4]$

Covariance:

$$\text{Cov}(X, X) = \frac{10}{4} = 2.5, \quad \text{Cov}(X, Y) = \frac{20}{4} = 5, \quad \text{Cov}(Y, Y) = \frac{40}{4} = 10$$

$$\mathbf{S} = \begin{bmatrix} 2.5 & 5 \\ 5 & 10 \end{bmatrix}$$

7.2 Solution to Problem 2

Characteristic equation:

$$(5 - \lambda)(2 - \lambda) - 4 = 0 \Rightarrow \lambda^2 - 7\lambda + 6 = 0$$

Eigenvalues: $\lambda_1 = 6, \lambda_2 = 1$

Eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{e}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$
$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

7.3 Solution to Problem 3

Total variance: $4.5 + 2.3 + 1.8 + 0.9 + 0.5 + 0.3 + 0.2 + 0.1 + 0.05 + 0.05 = 10.7$

Cumulative proportions:

- PC1: 42.06%
- PC1+PC2: 63.55%
- PC1+PC2+PC3: 80.37%
- PC1+PC2+PC3+PC4: 88.79%
- PC1+PC2+PC3+PC4+PC5: 93.46%

Need 5 components to exceed 90%.

7.4 Solution to Problem 4

Projection = dot product:

$$2 \times 0.6 + (-3) \times 0.8 = 1.2 - 2.4 = -1.2$$

7.5 Solution to Problem 5

1. Standardize features (different units/scales)
2. Compute covariance matrix to understand relationships
3. Perform PCA to identify principal components
4. Likely find:
 - PC1: "House size factor" (combines size and bedrooms)
 - PC2: "Location factor" (combines distance and school rating)
 - PC3: "Age factor"
5. Reduce to 2-3 dimensions for visualization/analysis
6. Interpret components to understand what drives house prices

8 Summary and References

8.1 Key Takeaways

Key Concept

- PCA is a **linear dimensionality reduction** technique
- It finds orthogonal directions of **maximum variance**
- Eigenvalues represent **variance magnitude**
- Eigenvectors represent **principal component directions**
- Choose components based on **variance explained**
- Always **standardize** when features have different scales

8.2 Mathematical Summary

Table 2: PCA Mathematical Summary

Concept	Mathematical Representation
Data Matrix	$\mathbf{X} \in \mathbb{R}^{N \times n}$
Mean Vector	$\bar{\mathbf{x}} = \frac{1}{N} \sum \mathbf{x}_i$
Centered Data	$\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}$
Covariance Matrix	$\mathbf{S} = \frac{1}{N-1} \mathbf{X}_c^T \mathbf{X}_c$
Eigen Decomposition	$\mathbf{S} \mathbf{v}_i = \lambda_i \mathbf{v}_i$
Principal Components	$\mathbf{Z} = \mathbf{X}_c \mathbf{V}_k$
Variance Explained	$\frac{\lambda_i}{\sum \lambda_j}$

8.3 Final Notes

- PCA is **unsupervised** - doesn't use target labels
- Works best when data has **linear relationships**

- Consider **kernel PCA** for nonlinear data
- Always **validate** results with domain knowledge
- PCA reduces dimensions but **not necessarily** improves prediction